

Algebra 1: Unit 5, Lesson 8

- Sharmina is baking cookies. She is able to put 18 cookies on the cookie sheet at a time. The number of cookies she bakes can be represented by the equation, $y = 18x$, where x is the number of batches of cookies she will need to make and y is the total number of cookies. Sharmina wants to make 100 cookies for the bake sale at her school. She represents this using the equation, $y = 100$.

Sharmina uses the systems of equations shown to determine the number of batches of cookies she will need to make.

$$y = 18x$$

$$y = 100$$

Complete the following statement.

To solve the system of equations, substitute 100 in for the y –variable in the equation, $y = 18x$. The number of batches of cookies that Sharmina needs to cook is _____.

- Lexi needs to rent a car for the weekend. Rent-A-Vehicle charges a \$100 rental fee plus \$0.20 per mile. Cars-a-Lot charges a \$75 rental fee plus \$0.35 per mile. Let $f(x)$ represent the cost to rent a car from Rent-A-Vehicle and $p(x)$ represent the cost to rent a car from Cars-a-Lot. What does $f(x) = p(x)$ represent?

- A linear equation is shown.

$$4x - 8y = 16$$

What is the equation rewritten in slope-intercept form by solving the equation for y ?